

The Hong Kong Polytechnic University
Subject Description Form

Subject Code	AP5002
Subject Title	Semiconductor Devices and Systems
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	To study the operational principles and underlying physical characteristics of silicon-based semiconductor electronic devices such as diodes (p - n , and metal-semiconductor junctions), and transistors (bipolar junction, field-effect) as well as their integrations.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) receive a deep understanding of the physical characteristics of p-n and metal-semiconductor junction diodes; (b) receive a sound understanding of the operating principles of the bipolar junction and field-effect transistors; (c) understand the formation of monolithic devices using bipolar, MOS, and CMOS technologies for the realization of integrated circuits; and (d) develop sufficient knowledge with which the students can later learn about newly developed nanometers scale microelectronics technologies.
Subject Synopsis/ Indicative Syllabus	Review of semiconductor physics: Crystal properties of semiconductors, energy bands, and charge carriers in semiconductors, excess carriers in semiconductors. Resistors, and capacitors: Diffused and implanted resistors, MOS capacitors. Diodes: Physical and electrical characteristics of p-n junction diodes and metal-semiconductor junction diodes. Transistors: Physical and electrical characteristics of bipolar junction transistors and field-effect transistors (e.g. JFET, MESFET, MOSFET). Monolithic devices: Design, fabrication, and characterization of bipolar transistors, merged transistors, MOS transistors, and CMOS transistors. Integrated circuits: Integration of other circuit elements on the same platform. The integration of resistors, capacitors, transistors via interconnections. Outlook: Recent progress and future development in semiconductor devices and their integration.
Teaching/Learning Methodology	Lecture: Operation principles of different silicon-based microelectronics devices will be explained. Concrete examples will be given to illustrate the difference between the different designs of microelectronics devices. Assignment sets will be given to assess the learning progress of students. Tutorial: Students will work on problem sets in the tutorials, which provide them opportunities to apply the knowledge gained in lectures.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Continuous Assessments (Assignments and tests)	40%	✓	✓	✓	✓
	2. Examination	60%	✓	✓	✓	✓
	Total	100 %				
Continuous assessments , which includes assignments and tests, aims at checking the progress of students throughout the course as well as assisting the students in self-monitoring their performance.						
The examination will use to assess the knowledge acquired by the students, as well as to determine the extent to which they have achieved the intended learning outcomes.						
Student Study Effort Expected	Class contact:					
	▪ Lecture				33 Hrs.	
	▪ Tutorial				6 Hrs	
	Other student study efforts:					
	▪ Reading and self-study				81 Hrs.	
	The total student study effort				120 Hrs.	
Reading List and References	S.M. Sze, Kwok K. Ng, Physics of Semiconductor Devices, John Wiley & Sons, 2006 (https://onlinelibrary.wiley.com/doi/book/10.1002/0470068329).					
	C.M. Hu, Modern Semiconductor Devices for Integrated Circuits, 1 st Edition, Pearson, March 22, 2009 (https://www.chu.berkeley.edu/modern-semiconductor-devices-for-integrated-circuits-chenming-calvin-hu-2010/).					
	J. A. Del Alamo, Integrated Microelectronic Devices: Physics and Modeling, Pearson, 2017 (http://www.materialdownload.in/article/Integrated-Microelectronic-Devices_367/).					
	W. Lambrechts, S. Sinha, J.A. Abdallah, J. Prinsloo, Extending Moore's Law through Advanced Semiconductor Design and Processing Techniques, CRC Press, 2018.					